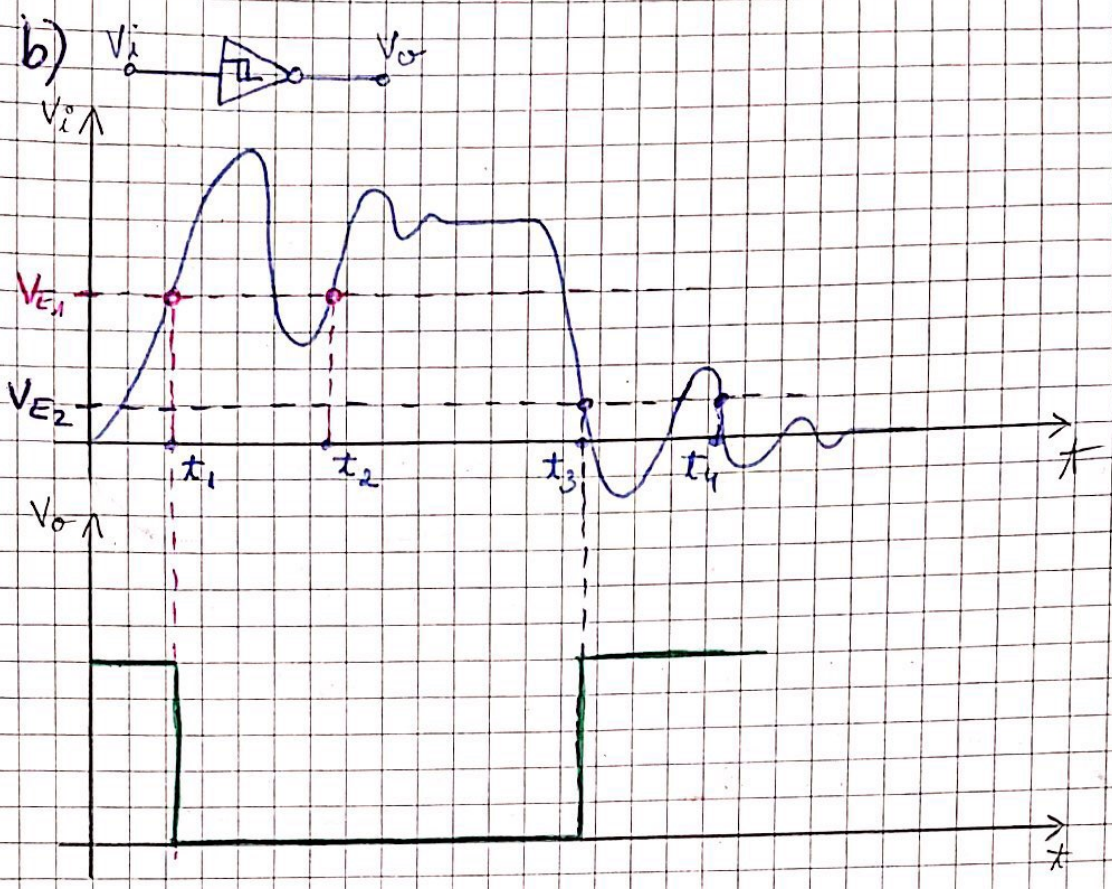
b)



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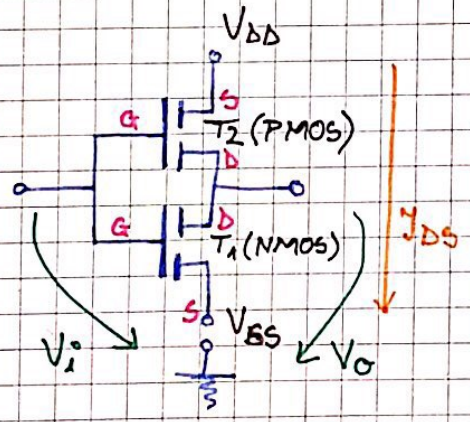
Familia de circuite integrate CMOS



$V_{DD} = 3 \div 18$ V, dar pt. exemplificare luăm: $V_{DD} = 5$ V

- Funcționare:

T_1 (NMOS) are tens. de prag: $V_{T_N} = V_T$

T_2 (PMOS) — " — : $V_{T_P} = -V_T$

$\Rightarrow V_{T_N} = |V_{T_P}| = V_T$

a) $V_i \leq V_T$: $\begin{cases} T_1 \text{ blocat} \Rightarrow J_{DS} = 0 \\ T_2 \text{ conduce } (|V_{T_P}| = |V_{DD} - V_i| > V_T) \Rightarrow V_o = V_{DD} - R_{T_P} \overset{0}{\overset{\shortparallel}{J_{DS}}} = \end{cases}$

$= V_{DD} = V_H$

Exemplu pt. rel din paranteză:

$V_T = \frac{V_{DD}}{2}$; $V_{DD} = 5$ V ; $V_i \leq 2{,}5$ V $\Rightarrow V_{T_N} \leq V_T = 2{,}5$ V

$|V_{T_N}| = |V_{DD} - V_i| = |5 - \text{cerva} \leq 2{,}5| \geq 2{,}5$ V $= |V_{T_P}|$

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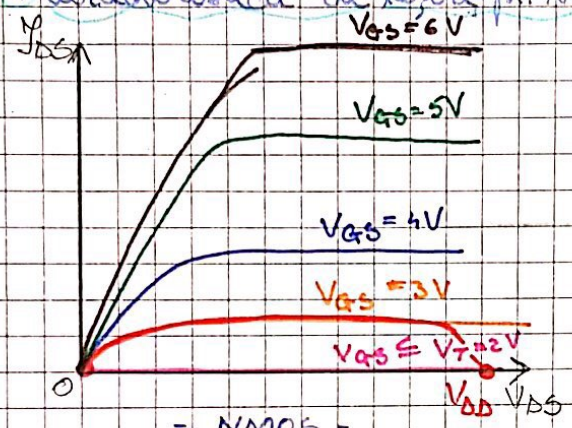
b) $V_i > V_T$: $\begin{cases} T_1 \text{ conduce} \Rightarrow V_o = V_{DS} + R_{TN} \cdot I_{DD} \cong V_{SS} = V_L \\ T_2 \text{ blocat} \Rightarrow I_{DS} = 0. \end{cases}$

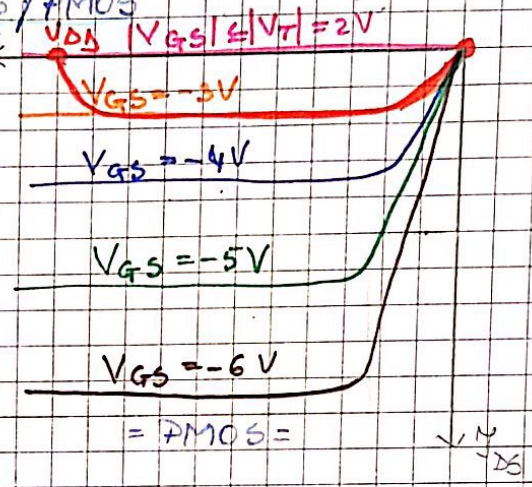
Parametrii de bază ai familiei CMOS

a) Nivelele de tensiune : se det. pe baza caracteristicii statice de transfer

- Caracteristica de ieșire pt. NMOS / PMOS



= NMOS =



= PMOS =

- Curba de sarcină pt. CMOS

a) pt. $V_i \leq V_T = 2V \Rightarrow I_{DS} = 0$ (T_1 blocat, T_2 conduce) $\Rightarrow$

$\Rightarrow V_o = V_{DD} - \underset{0}{V_{DS_2}} = V_{DD}$

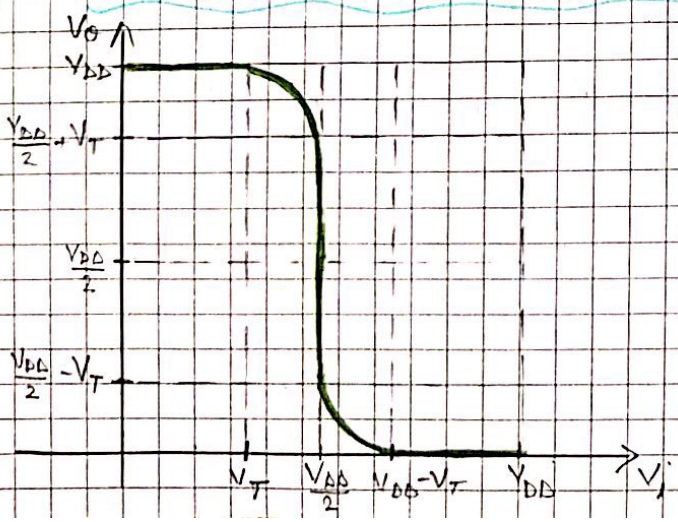
b) pt. $V_i = 3V \Rightarrow I_{DS} = 20\mu A$ (din alt curs) $= I_{DS_1} = I_{DS_2} \Rightarrow$

$\Rightarrow V_o = V_{DD} - V_{DS_2} = 6 - 3 = 3V$
↳ ales de noi

? c) $V_i \geq \cancel{3}\,4V \Rightarrow T_2$ blocat $\Rightarrow I_{DS} = 0 \Rightarrow V_o = 0$: $V_{DS_1} = 0$; $V_{DS_2} = 6V$

$\Rightarrow$ Curbele de sarcină a NMOS și PMOS sunt

- Caracteristica statică de transfer a CMOS ($V_o = f(V_i)$)



V_T – tens. de prag pt. care NMOS se deschide

$V_{DD} - V_T$ – tens. de prag pt. care PMOS se blochează

a) pt. $0 \leq V_i \leq V_T \Rightarrow T_1$ blocat ($I_{DS} = 0$);

T_2 conduce

$V_o = V_{DD} - R_{TN} \cdot I_{DS} = V_{DD}$

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b) pt. $V_{DD} - V_T \le V_i \le V_{DD} \Rightarrow T_2$ blocat $\Rightarrow I_{DS} = 0 \Rightarrow V_o = V_{SS} + I_{DS} \cdot R_{TN}$

$\Rightarrow V_o = V_{SS} = 0\,V$

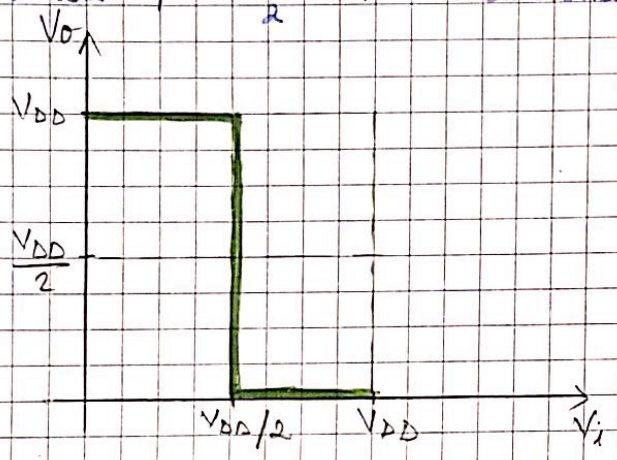
c) pt. $V_i = V_{DD}/2 \Rightarrow T_1$ și T_2 conduc $\Rightarrow V_o = \frac{V_{DD}}{2}$

d) pt. $V_T < V_i < \frac{V_{DD}}{2} \Rightarrow T_1$ începe să conducă

e) pt. $V_{DD}/2 < V_i < V_{DD} - V_T \Rightarrow T_2$ începe să se blocheze $\Rightarrow$

$\Rightarrow I_{DS}$ începe să scadă $\Rightarrow V_o$ tinde să scadă

- Dacă $V_T = \frac{V_{DD}}{2} \Rightarrow$ Caracteristica statică de transfer ideală

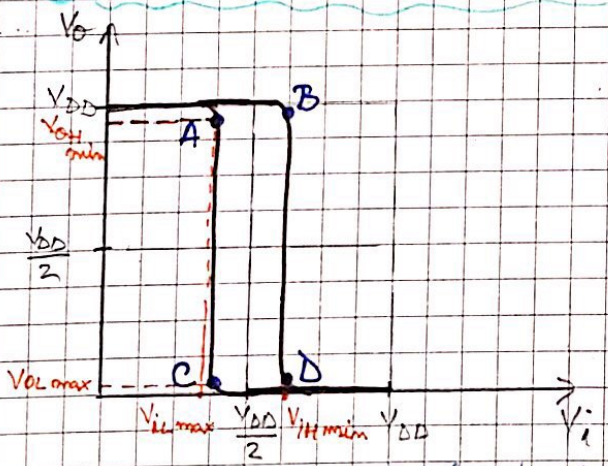


Știm că, în practică se permite:

$\frac{V_{DD}}{3} \le V_T \le \frac{2}{3} V_{DD} \Rightarrow$

$\Rightarrow$ Caracteristicile statice de transfer aparțin unor intervale

- Caracteristica statică de transfer limită; pe baza ei se definesc nivelele de tensiune



- Se trasează o tg. în punctele în care $\frac{\Delta V_o}{\Delta V_i} = 1 \Rightarrow$ Rezultă punctele A, B, C, D din care se consideră situația c.m. defavorabilă (A și D)
- $V_{OH\,min} \approx V_{DD}$ la ieșire; $V_{OL\,max} \approx 0\,V \Rightarrow$

$\Rightarrow$ CMOS are nivele de tensiune aproape ideale

- $V_{IL\,max} = \frac{1}{3} V_{DD}$; $V_{IH\,min} = \frac{2}{3} V_{DD} \Rightarrow$ Și ele tind către $\frac{V_{DD}}{2}$ (ideal)
- Deci CMOS ROCKS!

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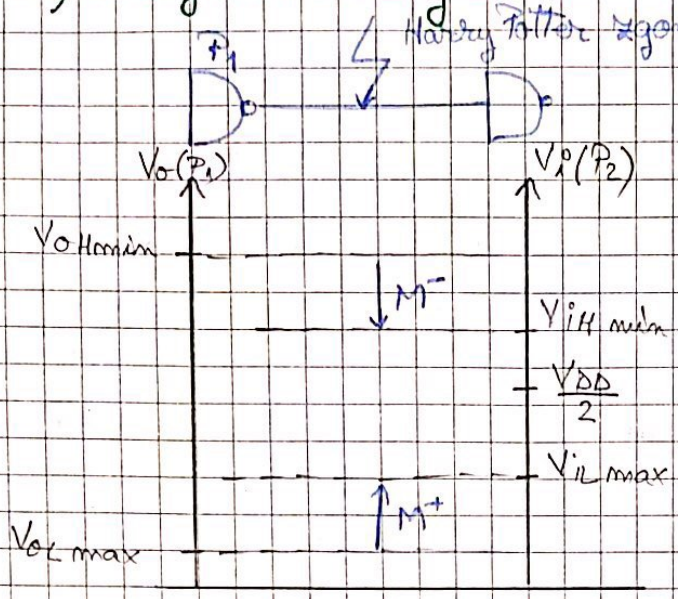
b) Marginea de zgomot



$$M^- = V_{OH\,min} - V_{iH\,min} \approx \frac{1}{3} V_{DD}$$

$$M^+ = V_{iL\,max} - V_{OL\,max} \approx \frac{1}{3} V_{DD}$$

- Exemplu numeric:
  - La TTL: $M^+ = M^- = 0{,}4\,V$
  - La CMOS pt $V_{DD} = 5\,V$:

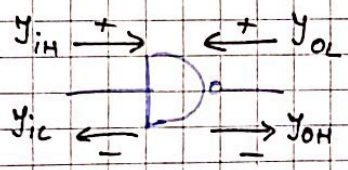
    $M^+ = M^- = 1{,}66\,V \approx \frac{5}{3}$ (mai bune ca la TTL)

c) Factorul de încărcare

$$FE_H = \left|\frac{I_{OH}}{I_{iH}}\right| \in \mathbb{Z}_+$$

$$FE_L = \left|\frac{I_{OL}}{I_{iL}}\right| \in \mathbb{Z}_+$$

factor de încărcare la ieșire pt. fiecare nivel



- $\forall$ intră e cu (+)
- $\forall$ iese e cu (−)

$\Rightarrow |-I_{iL}| = I_{iH} \approx$ ~~0 μA~~

$|-I_{OH}| = I_{OL} =$ sute de μA ÷ sute de mA

$\Rightarrow FE_H, FE_L \longrightarrow \infty$

d) Puterea totală

$$P_{tot} = \underbrace{P_{DD}}_{\text{statică}} + \underbrace{P_D}_{\text{dinamică}} + \underbrace{P_{DS}}_{\text{dinamică suplimentară}}$$

a. $P_{DD} = \frac{I_{DSH} + I_{DSL}}{2} V_{DD}$

$I_{DSH} = I_{DSL} = 0$

ex: $I_{DD} = 10\,mA$ pt $V_{DD} = 5\,V \Rightarrow P_{DD} = \frac{10+10}{2} \cdot 10^{-3} \cdot 5 = 50\,mW$

(La TTL: 10 mW)

b. $P_D = C_p \cdot F \cdot V_{DD}^2$ (F – frecvența)

ex: TTL + CMOS:

$C_p = 50\,pF$; $F = 1\,MHz$

$V_{CC} = V_{DD} = 5\,V$

$\Rightarrow P_D = 1{,}25\,mW$

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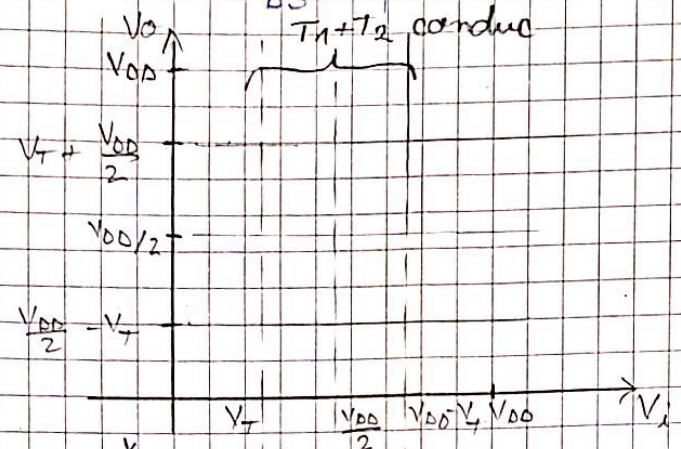
c. P_{DS} depinde de cum arată caract. statică de transfer.

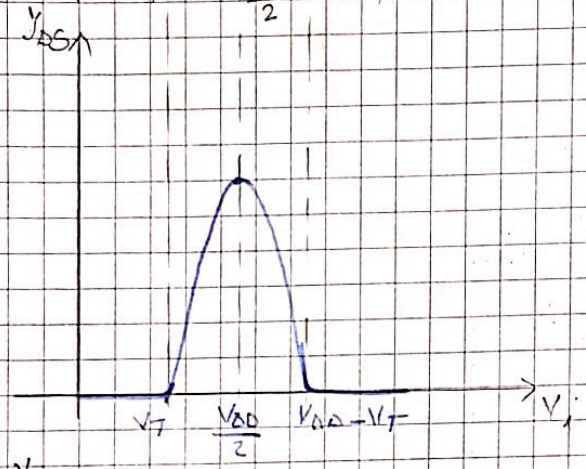


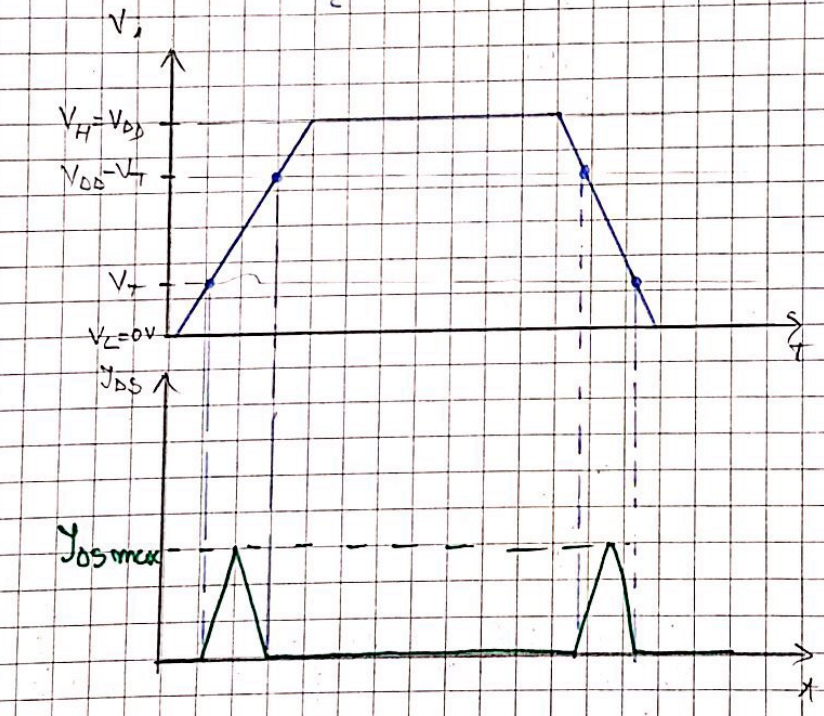
Scenariu: T_1 și T_2 conduc $\Rightarrow$

$$\Rightarrow I_{DS} = \frac{V_{DD}}{R_{T_N}+R_{T_P}} = \frac{5}{(1+1)10^3} =$$

$= 2{,}5$ mA pt. că

$R_{T_N} = R_{T_P} = 1\,k\Omega$





$$P_{DS} = \frac{I_{DS\,max}}{2}\left(V_{DD}-V_T-2\right)\cdot\frac{t_1+t_2}{T} =$$

$$= \frac{I_{DS\,max}}{2}\left(V_{DD}-2V_T\right)\cdot f\cdot(t_1+t_2)$$

Deci dacă $V_T = \frac{V_{DD}}{2} \Rightarrow$

$\Rightarrow P_{DS} = 0.$

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